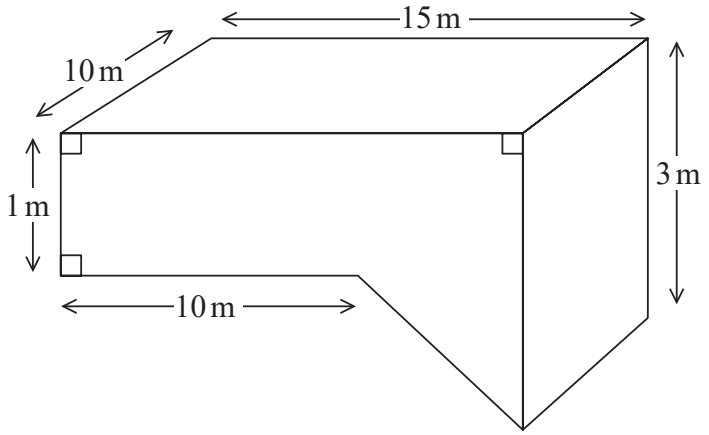
4



The diagram shows a swimming pool.
The swimming pool is in the shape of a prism.
The swimming pool is filled with water at a rate of 5 litres per second.

Jeremy has 10 hours to fill the swimming pool.
$1\,m^3 = 1000$ litres.

Will he completely fill the swimming pool in 10 hours?
You must show all your working.

Cross-section = rectangle + triangle

= 15 × 1 + ½ × 5 × 2 = 20 m².

1 mark
Cross-section20 m²

Pool volume = 20 × 10 = 200 m³.

1 mark
Volume200 m³

Water needed = 200 × 1000 = 200000 litres.

1 mark
Convert to litres

Time = 200000 ÷ 5 = 40000 seconds.

1 mark
40000 seconds

10 hours = 10 × 60 × 60 = 36000 seconds.

No: 40000 > 36000, so more than10 hours is needed.

1 mark
Compare same units

(Total for Question 4 is 5 marks)